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Smt. Kamalaben Gambhirchand Shah Department of Computer Applications

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S.N.D. T Women's University 338, R.A.Kidwai Road
Matunga, Mumbai

A REPORT ON

Introduction to Statistical Methods Project- Statistical Analysis of Government Open Data using Computer Applications

CLASS:SYBCA

AUTHORS & AFFILIATION:

DISHA THAKUR (ROLL NO :164)

REETIKA SHARMA (ROLL NO :157)

I. INTRODUCTION & OBJECTIVES

Crime statistics offer important insights into society's dynamics, public safety, and the effectiveness of law enforcement. In the present era, statistical methods are widely used to analyze and interpret real-world data in a systematic and meaningful manner. This project focuses on the analysis of crime data for the year 2019, specifically covering cases reported, cases disposed, and gender-wise arrests under the Indian Penal Code (IPC) in metropolitan cities. Such data helps in understanding how crimes are recorded, processed, and handled by authorities.

Statistical techniques are applied to identify patterns, relationships, and significant differences in the data. In particular, the study examines the relationship between reported cases and arrests, and analyzes differences between male and female arrest patterns. Overall, this analysis helps in transforming raw data into meaningful insights, thereby contributing to a better understanding of crime trends and supporting data-driven decision making.

This project also emphasizes the practical application of statistical concepts learned theoretically in the subject. It provides hands-on experience in analyzing real-world data using appropriate statistical techniques. This approach helps in better understanding and strengthening conceptual knowledge through practical implementation.

Objectives:

- To analyze crime data using statistical methods.
- To calculate measures of central tendency such as mean and median.
- To study variability using standard deviation and variance.
- To identify relationships between variables using correlation analysis.
- To apply regression analysis to understand dependency between variables.
- To perform hypothesis testing using t-test to compare male and female arrests.
- To validate and verify results using Python.

II. DATASET DESCRIPTION

Two datasets from data.gov.in for the year 2019 were selected:

- Dataset A : City, Age Group and Gender-wise Persons Arrested under IPC Crimes in Metropolitan Cities during 2019.

This dataset provides information about the number of persons arrested in different metropolitan cities. It shows the number of male and female arrests along with the total arrests in each city. This helps in comparing crime-related data across different locations.

VARIABLES USED FOR ANALYSIS:

1.Total male arrested 2.Total female arrested 3. Total persons arrested

S. No	City	Total - Male	Total - Female	Total - Arrested
1	Ahmedabad (Gujarat)	31898	3054	34952
2	Bengaluru(Karnataka)	26337	2021	28358
3	Chennai(Tamil Nadu)	53804	4704	58508
4	Coimbatore(Tamil Nadu)	6940	1032	7972
5	Delhi	96959	2127	99086
6	Ghaziabad(Uttar Pradesh)	4270	7	4277
7	Hyderabad(Telangana)	7890	310	8200
8	Indore(Madhya Pradesh)	9231	537	9768
9	Jaipur(Rajasthan)	15109	1235	16344
10	Kanpur(Uttar Pradesh)	8349	76	8425
11	Kochi(Kerala)	14930	279	15209
12	Kolkata(West Bengal)	12249	2243	14492
13	Kozhikode(Kerala)	4087	171	4258
14	Lucknow(Uttar Pradesh)	8272	0	8272
15	Mumbai(Maharashtra)	51312	3231	54543
16	Nagpur(Maharashtra)	10614	464	11078
17	Patna(Bihar)	8236	48	8284
18	Pune(Maharashtra)	15985	1138	17123
19	Surat(Gujarat)	24378	1242	25620
Grand Total	TOTAL CITIES	410850	23919	434769

- Dataset B : Crime Head-wise Police Disposal of IPC Crime Cases during 2019.

This dataset contains detailed information about different types of crimes and the number of cases reported and disposed during the year 2019. It includes various crime categories such as murder, theft, kidnapping, assault, and others. Some categories are further divided into subcategories, giving a more detailed view of the data.

VARIABLES USED FOR ANALYSIS:

1. Total cases reported during the year
2. Total cases disposed off during the year

S. No	Crime Head	Total Cases Reported	Total Cases Disposed Off
1	Murder	28918	27616
2	Culpable Homicide not amounting to Murder	3470	3315
3	Causing Death by Negligence	144842	140278
3.1	Deaths due to Negligence relating to Road Accidents	136587	132229
3.1.1	Hit and Run	47530	45129
3.1.2	Other Accidents	89057	87100
3.2	Deaths due to Negligence relating to Rail Accidents	55	39
3.3	Deaths due to Medical Negligence	210	212
3.4	Deaths due to Negligence of Civic Bodies	147	125
3.5	Deaths due to other Negligence	7843	7673
4	Dowry Deaths	7115	6964
5	Abetment of Suicide	8315	7748
6	Attempt to Commit Murder	51254	50280
7	Attempt to commit Culpable Homicide	7766	7763
8	Attempt to Commit Suicide	1637	1601
9	Miscarriage, Infanticide, Foeticide and Abandonment	1950	1810
10	Hurt	545061	527300
10.1	Simple Hurt	455946	438061
10.1.1	Voluntarily Causing Simple Hurt	261047	251612
10.1.2	Voluntarily causing hurt by Dangerous weapons or means	162907	155108
10.1.3	Voluntarily Causing hurt/Force to deter public servant from his duty	16781	16409
10.1.4	Causing Hurt by act endangering life/safety of others	9496	9306
10.1.5	Other Simple Hurt	5715	5626
10.2	Grievous Hurt	89115	89239
10.2.1	Voluntarily causing grievous hurt	61685	61832
10.2.2	Voluntarily causing grievous hurt by dangerous weapons or means	20373	21189
10.2.3	Acid Attack	240	236
10.2.4	Attempt to Acid Attack	60	51
10.2.5	Voluntarily Causing Grievous hurt/Force to deter public servant from his duty	1742	1658
10.2.6	Causing Grievous Hurt by act endangering life/safety of others	3979	3411
10.2.7	Other Grievous Hurt	1036	862
11	Wrongful Restraint/Confinement	18683	18485
12	Assault on Women with Intent to Outrage her Modesty	88367	87013
12.1	Assault on Women with Intent to Outrage her Modesty	48586	48950
12.2	Sexual Harassment	18334	18069
12.2.1	at Work or Office Premises	505	534
12.2.2	In Public Transport System	606	611
12.2.3	In Shelter Homes for women and Children	473	425
12.2.4	Other Places	16750	16499
12.3	Assault or use of Criminal Force on women with intent to Disrobe	11238	10425
12.4	Voyeurism	1319	1185
12.5	Stalking	8890	8384
13	Kidnapping and Abduction	105037	101912
13.1	Kidnapping and Abduction	56591	53337
13.1.1	Missing Children Deemed as Kidnapped	28370	25749
13.1.2	Other Kidnapping & Abduction	28221	27588

13.2	Kidnapping and Abduction for the Purpose of Begging	42	36
13.3	Kidnapping and Abduction in order to Murder	649	633
13.4	Kidnapping for Ransom	580	602
13.5	Kidnapping and Abduction of Women to compel her for marriage	32260	32676
13.6	Procurator of Minor Girls	3117	2833
13.7	Importation of Girls from Foreign Country	3	1
13.8	Other Kidnapping & Abduction	11795	11794
14	Human Trafficking	1334	1189
15	Exploitation of Trafficked Person	183	157
16	Selling of Minors for Prostitution	24	42
17	Buying of Minors for Prostitution	8	6
18	Rape	32033	30470
19	Attempt to Commit Rape	3944	4036
20	Unnatural Offences	1004	1100
Total	Offences Affecting the Human Body (Total)	1050945	1019085
21	Offences against State	166	160
21.1	Sedition	93	70
21.2	Others	73	90
Total	Offences against the State (Total)	166	160
22	Unlawful Assembly	8889	8787
23	Riots	46209	49235
23.1	Communal/Religious	440	486
	Sectarian	118	120
23.3	Industrial	273	320
23.4	Political	1209	1285
23.5	Caste Conflict	492	503
23.6	Agrarian	1579	1836
23.7	Students	391	417
23.8	Money Dispute	961	979
23.9	Water Dispute	793	697
23.1	Land/Property Dispute	8880	10016
23.11	Family Disputes	3479	3788
23.12	Enmity/Rivalry	4431	4157
23.13	Electricity/Power Supply Disputes	37	36
23.14	Rioting while in Aandolan/Morcha	1442	1656
23.15	Rioting/Attacks on Police Personnel or Govt Servants	1044	1328
23.16	Other Riots	20640	21611
24	Offences promoting enmity between different groups	1113	861
24.1	Promoting enmity between groups	1055	805
24.2	Imputation, assertions prejudicial to national integration	58	56
25	Affray	7148	8017
Total	Offences against Public Tranquillity (Total)	63359	66900
26	Theft	675916	681040
26.1	Auto/Motor Vehicle Theft	237884	236302
26.2	Other Thefts	438032	444738
27	Burglary	100897	98655
27.1	Day Time	14805	14861
27.2	Night	86092	83794

28	Extortion & Blackmailing	11067	10191
29	Robbery	31065	29478
30	Attempt to Commit Dacoity/Robbery	596	543
31	Dacoity	3176	3463
31.1	Dacoity	3132	3414
31.2	Dacoity with Murder	44	49
32	Making Preparation and Assembly for committing Dacoity	3338	3449
33	Criminal Misappropriation	497	400
34	Criminal Breach of Trust	20833	20349
35	Dishonestly Receiving/Dealing-in Stolen Property	7233	7301
Total	Offences against Property (Total)	854618	854869
36	Counterfeiting	1040	1138
36.1	Counterfeit Coin	20	19
36.2	Counterfeiting Government Stamp	7	5
36.3	Counterfeiting of Seal or Mark	3	3
36.4	Counterfeit Currency & Bank notes	1010	1111
37	Forgery, Cheating & Fraud	143909	128633
37.1	Fraud	23383	19608
37.1.1	Bank	2271	1536
37.1.2	ATMs	3692	3033
37.1.3	Credit Card/Debit Card	1822	1354
37.1.4	Other Frauds	15598	13685
37.2	Other Cheating	104332	93599
37.3	Other Forgery	16194	15426
Total	Offences Relating to Documents & Property Marks (Total)	144949	129771
38	Offences relating to Elections	5545	5036
39	Disobedience to order duly promulgated by Public Servant	29496	27039
40	Harbouring an Offender	81	81
41	Offences relating to Adulteration or Sale of Food/Drugs	2959	2608
42	Rash Driving on Public way	417024	410357
42.1	Rash Driving on Public Way	144059	142071
42.2	Causing Hurt by Rash/Negligent Driving on Public Way	160794	157438
42.3	Causing Grievous Hurt by Rash/Negligent Driving on Public	112171	110848
43	Obstruction on Public way	93343	93819
44	Sale of obscene Books/Objects	108	88
45	Obscene Acts and Songs at Public Places	24708	24590
46	Offences relating to Religion	1459	1339
47	Cheating by Impersonation	775	751
48	Offences related to Mischief	2679	2423
49	Arson	8420	8968
50	Criminal Trespass	39477	39325
51	Cruelty by Husband or his Relatives	125298	118487
52	Circulate False/Fake News/Rumours	476	347
53	Criminal Intimidation	101422	98619
54	Insult to the Modesty of Women	6939	6863
Total	Miscellaneous IPC Crimes(Total)	860209	840740
55	Other IPC crimes	251455	248798
Grand Total	Total Cognizable IPC crimes	3225701	3160323

III. TOOLS USED

- Python was used as the primary programming language for data analysis due to its simplicity and powerful libraries.

The following libraries were used:

- pandas: for data loading, cleaning, and manipulation
 - numpy: for performing numerical calculations
 - scipy (stats module): for conducting hypothesis testing, specifically the t-test.
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- Chartgo: A web-based data visualization tool used to create clear and presentable graphical representations of the computed results. The summarized output data (such as totals, averages, and comparisons between male and female arrests) was manually entered into ChartGo to generate bar charts, line graphs, and histograms. This tool was particularly useful for producing neat and customizable charts suitable for inclusion in the final report.
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- In addition, basic verification of results performed using Microsoft Excel, which helps in cross-checking calculations such as mean and median.

These tools enable accurate computation and effective visualization of the dataset.

IV. STATISTICAL ANALYSIS & CALCULATIONS

Various statistical techniques were applied to analyze the dataset:

1) Measures of Central Tendency -

The mean and median were calculated to understand the average and central values of:

- Reported cases
- Disposed cases
- Total persons arrested

2) Quartiles and Percentiles

Quartiles and percentiles were calculated to understand the distribution of the data and identify how values are spread across different ranges.

These measures were calculated for:

- Reported cases
- Total persons arrested

3) Measures of Dispersion

Variance and Standard Deviation for total persons arrested and total cases reported were calculated to understand how much the data varies from the mean.

4) Correlation Analysis

Correlation was used to measure the strength and direction of the relationship between variables

For :

- Relationship between reported cases and disposed cases
- Relationship between male and female arrests

5) Regression Analysis

Linear regression was applied to study how one variable depends on another

For:

- Independent variable (X): Reported cases, Dependent variable (Y): Disposed Cases
- Independent variable (X): Male arrests, Dependent variable (Y): Female arrests

6) Hypothesis Testing (T-Test)

A t-test was conducted to compare male and female arrests

- Null Hypothesis (H_0): There is no significant difference between the average number of male and female arrests.
- Alternative Hypothesis (H_1): There is a significant difference between the average number of male and female arrests.
- Significance Level: 0.05 (5%)

□ PYTHON CODE FOR ANALYSIS OF VARIABLES - TOTAL ARRESTS, MALE ARRESTS AND FEMALE ARRESTS:

```
import pandas as pd

import numpy as np

from scipy import stats

# Load dataset

data = pd.read_csv("City, Age Group and Gender-wise Persons Arrested under
IPC Crimes in Metropolitan Cities during 2019.csv")

data=data[:-1] #removing the last total row for calculations

#columns

total = 'Total - Arrested'

male = 'Total - Male'

female = 'Total - Female'

# 1. MEAN, MEDIAN

print("Mean of total persons arrested:", data[total].mean())

print("Median of total persons arrested:", data[total].median())

# 2. QUANTILES, PERCENTILES

print("\nQUANTILES of total persons arrested:-")

print("1st quartile:", data[total].quantile(0.25))

print("2nd quartile:", data[total].quantile(0.5))

print("3rd quartile:", data[total].quantile(0.75))

print("\nPERCENTILES of total persons arrested:-")

print("25th Percentile:", data[total].quantile(0.25))

print("50th Percentile:", data[total].quantile(0.50))

print("75th Percentile:", data[total].quantile(0.75))
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print("90th Percentile:", data[total].quantile(0.90))

# 3. STANDARD DEVIATION & VARIANCE

print("\nStandard Deviation of total persons arrested:", data[total].std())

print("Variance of total persons arrested:", data[total].var())

# 4. CORRELATION

print("\nCorrelation between male and female arrests:")

print(data['Total    - Male'].corr(data['Total    - Female']))

# 5. REGRESSION

print("\nREGRESSION (female vs male)")

x = data[male]

y = data[female]

slope, intercept = np.polyfit(x, y, 1)

print("Slope:", slope)

print("Intercept:", intercept)

print("Regression equation: Female arrests=",intercept ,"+", slope ,"Male
arrests")

# 6. T-Test (two sample test)

t_stat, p_value = stats.ttest_ind(data[male], data[female])

print("\nT-TEST:-")

print("T-Statistic:", t_stat)

print("P-Value:", p_value)

if p_value < 0.05:

    print("Result: Since P-value is LESS than 0.05, the difference is
statistically significant.")

else:

    print("Result: Since P-value is GREATER than 0.05, the difference is not
significant.")

```

□ PYTHON CODE FOR ANALYSIS OF VARIABLES - TOTAL CASES REPORTED AND TOTAL CASES DISPOSED OFF DURING THE YEAR:

```
import pandas as pd

import numpy as np

# Load dataset

data = pd.read_csv("Crime Head-wise Police Disposal of IPC Crime Cases (Crime
Head-wise) during 2019.csv")

data=data[:-1] #removing the last total row for calculations

# Select columns

reported = data['Reported']

disposed = data['Disposed']

# 1. MEAN & MEDIAN

print("MEAN:")

print("Reported cases:", reported.mean())

print("Disposed cases:", disposed.mean())

print("\nMEDIAN:")

print("Reported cases:", reported.median())

print("Disposed cases:", disposed.median())

# 2. QUANTILES & PERCENTILES

print("\nQUANTILES (Reported cases):")

print("Q1:", reported.quantile(0.25))

print("Q2:", reported.quantile(0.50))

print("Q3:", reported.quantile(0.75))

print("\nPERCENTILES (Reported cases):")

print("25%:", reported.quantile(0.25))
```

```

print("50%:", reported.quantile(0.50))

print("75%:", reported.quantile(0.75))

print("90%:", reported.quantile(0.90))

# 3. STANDARD DEVIATION & VARIANCE

print("\nStandard Deviation:", reported.std())

print("Variance:", reported.var())

# 4. CORRELATION

print("\nCorrelation (Reported vs Disposed cases):")

print(reported.corr(disposed))

# 5. REGRESSION

print("\n REGRESSION")

x = reported

y = disposed

slope, intercept = np.polyfit(x, y, 1)

print("Slope:", slope)

print("Intercept:", intercept)

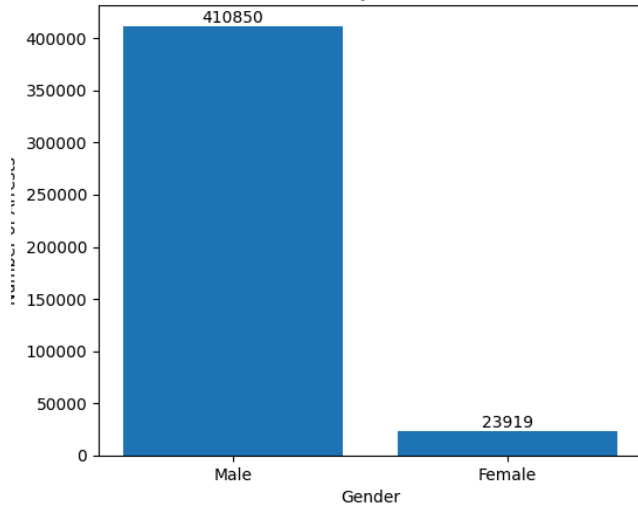
print("Regression equation: total disposed cases during the year =",intercept
, "+", slope , "total reported cases during the year")

```

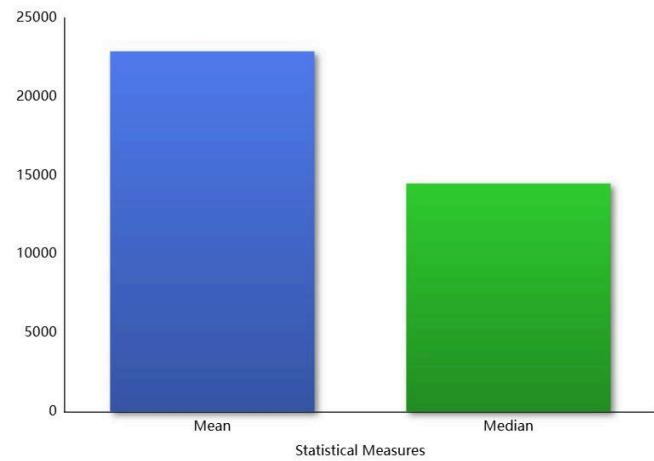
V. GRAPHS, TABLES & CHARTS

DATASET A

Total Arrests by Gender (2019)



Comparison of Mean and Median of Total Arrests (2019)



VALUE

Figure 1: Bar graph showing total arrests by gender (male and female) in 2019.

Figure 2: Bar graph comparing the mean and median of total arrests.

City-wise Total Arrests (2019)

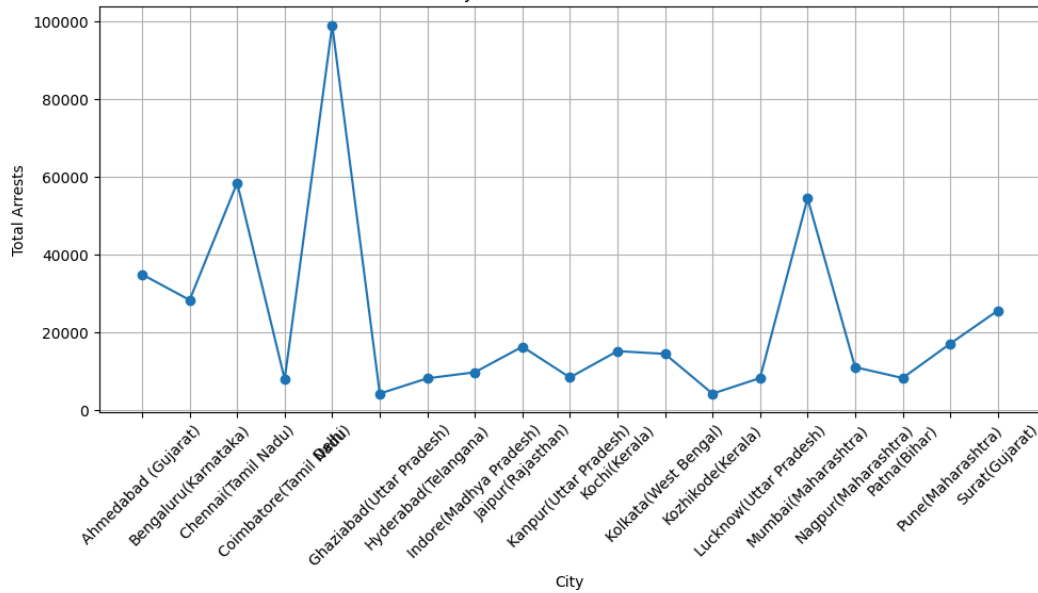


Figure 3: Line graph illustrating city-wise total arrests in 2019.

DATASET B

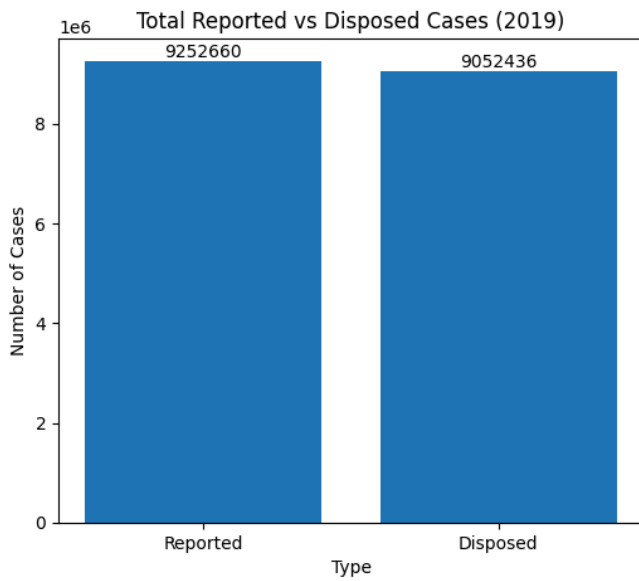


Figure 4: Bar graph comparing total reported and disposed cases in 2019.

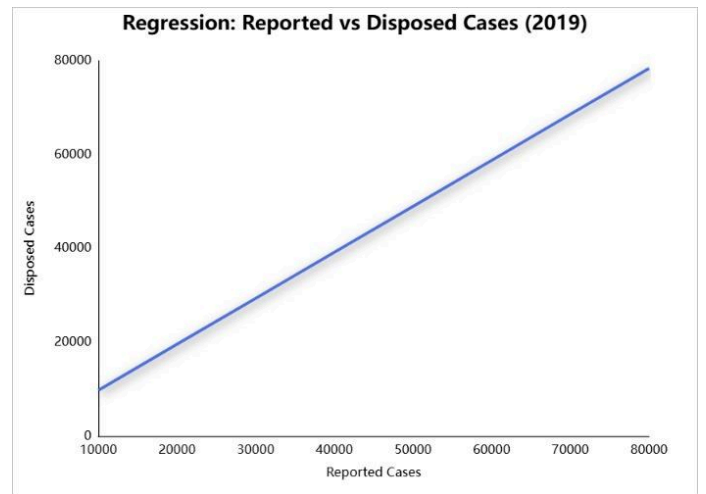


Figure 5: Graph showing the relationship between reported and disposed cases using regression analysis (2019).

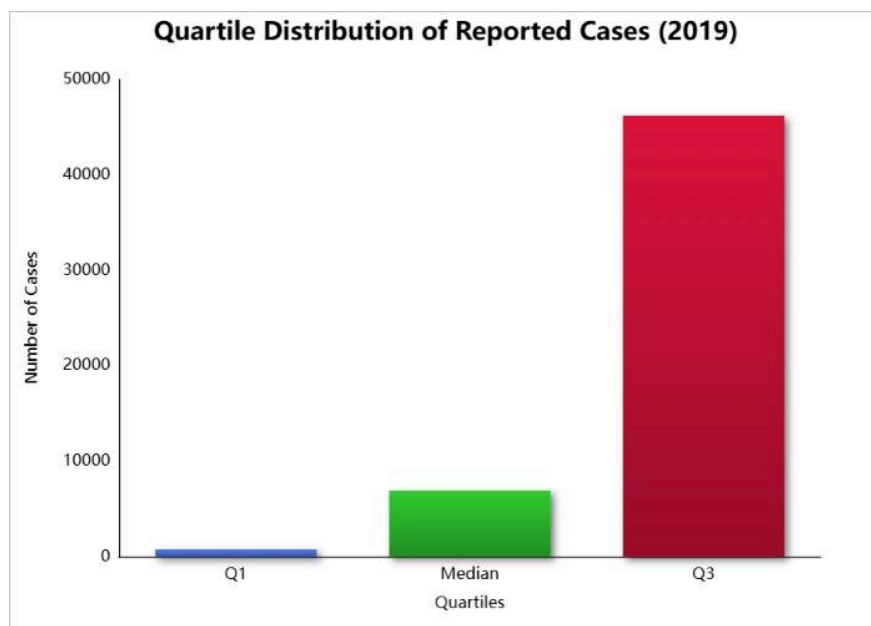


Figure 6: Bar graph representing the quartile distribution of reported cases in 2019.

VI. RESULTS & INTERPRETATION

☐ STATISTICAL RESULTS FOR DATASET A

Metric	Value
Mean (Total Arrests)	22882.58
Median (Total Arrests)	14492.00
Standard Deviation (Total Arrests)	24203.73
Variance (Total Arrests)	585820315.04
1st Quartile (Q1)	8278.00
2nd Quartile (Q2)	14492.00
3rd Quartile (Q3)	26989.00
25th Percentile(P25)	8278.0
50th Percentile(P50)	14492.0
75th percentile(P75)	26989.0
90th Percentile(P90)	55336.0
Correlation (Male vs Female)	0.6822
T-Statistic	3.8067
P-Value	0.0005

Regression Equation: Female arrests = 419.85 + 0.0388 * Male arrests

☐ STATISTICAL RESULTS FOR DATASET B

Metric	Value
Mean (Reported Cases)	65621.70
Mean (Disposed Cases)	64201.67
Median (Reported Cases)	6939.00
Median (Disposed Cases)	6863.00
Standard Deviation (Reported Cases)	164367.58
Variance (Reported Cases)	27016702254.27
Correlation (Reported vs Disposed)	0.9998
1st Quartile (Q1)	775.0
2nd Quartile (Q2)	6939.0
3rd Quartile (Q3)	46209.0
25th Percentile(P25)	775.0
50th Percentile(P50)	6939.0
75th percentile(P75)	46209.0
90th Percentile(P90)	144842.0

Regression Equation : Total disposed cases = -272.40 + 0.9825 * Total reported cases

- The analysis of Dataset A shows a strong positive correlation of 0.6822 between male and female arrests. This suggests that changes in crime levels affect both genders similarly in metropolitan cities. However, the number of male arrests is consistently higher than that of female arrests. This indicates a difference between the genders in crime involvement or law enforcement patterns.

The T-test gave a P-value of 0.0005. At a 5% significance level, we reject the null hypothesis. This confirms that the difference between the average number of male and female arrests is statistically significant. It means that the observed difference is not due to random chance and points to a real pattern in the data.

Furthermore, measures of central tendency and dispersion show that male arrests have a higher mean and more variation compared to female arrests. This suggests that males are involved in crimes more often and across a wider range of cities.

- In Dataset B, the near-perfect correlation of 0.9998 between reported and disposed cases shows an almost direct link between crime reporting and case resolution. This means that as the number of reported cases rises, the number of disposed cases also increases at a nearly equal rate.

This strong correlation indicates good consistency and efficiency in handling cases across different crime types. However, it doesn't mean all cases are fully resolved since disposal may include pending closures or other outcomes.

- Overall, the graphical representations support these findings by clearly showing trends, distributions, and comparisons between variables.

VII. CONCLUSION

In this project, we successfully used several statistical techniques, including measures of central tendency, dispersion, correlation analysis, and hypothesis testing, to examine two datasets related to IPC crimes in 2019.

The study found a significant difference between male and female arrest patterns, showing gender-based disparities in crime involvement. Additionally, the strong correlation in Dataset B points to a consistent relationship between reported and disposed cases, indicating a systematic approach to handling criminal cases.

The combination of Python-based analysis with visualization tools allowed for precise calculations and clear presentation of results. Using graphical methods improved the understanding of the statistical findings.

However, this analysis is limited to one year and specific metropolitan areas, which may not reflect long-term trends or rural crime patterns. Future studies could include data from multiple years and more variables to give a clearer picture.

Overall, this project offers data-driven insights into crime trends and law enforcement effectiveness. This project helped us gain practical knowledge by applying the theoretical concepts learned in our subject to real-world data analysis.

VIII. REFERENCES

- DATASETS from - data.gov.in

DatasetA:

<https://www.data.gov.in/resource/city-age-group-and-gender-wise-persons-arrested-under-ipc-crimes-metropolitan-cities-0>

Dataset B:

<https://www.data.gov.in/resource/crime-head-wise-police-disposal-ipc-crime-cases-crime-head-wise-during-2019>